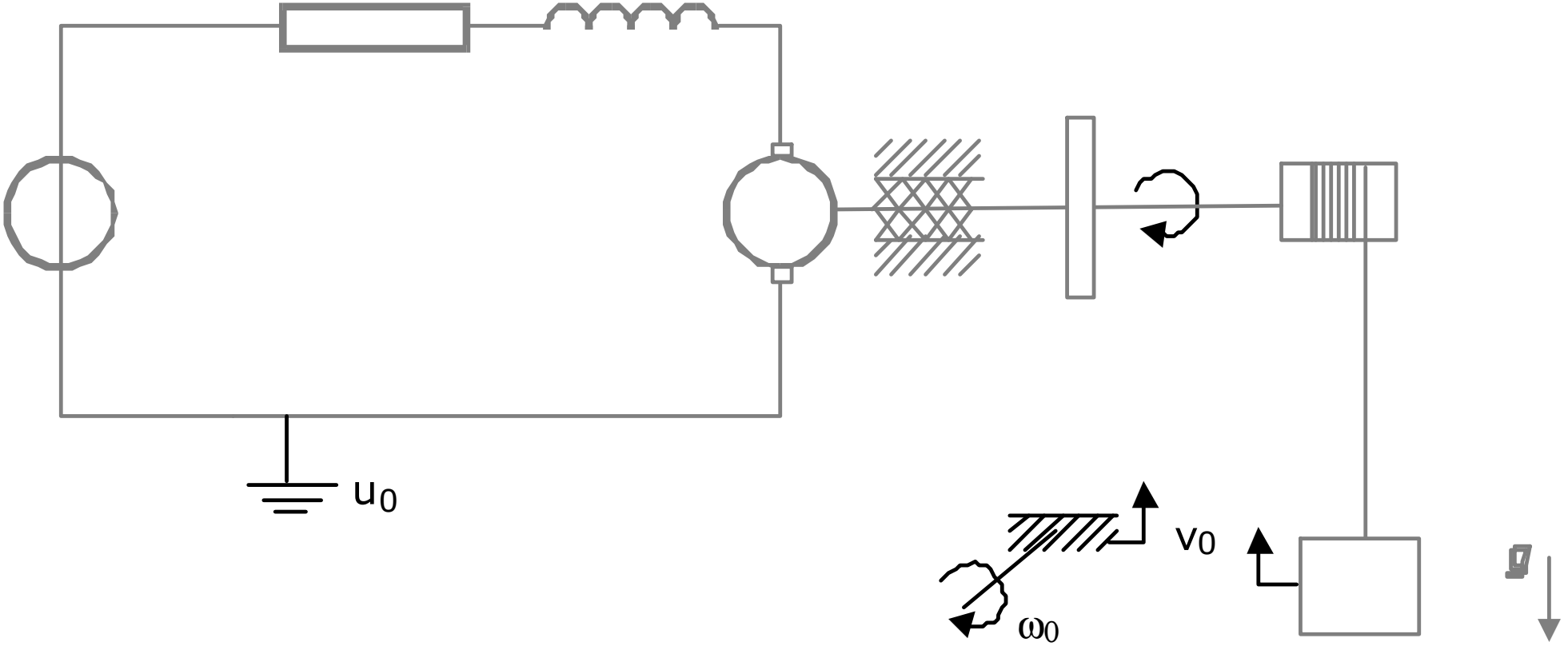

u_0
v_0
ω_0